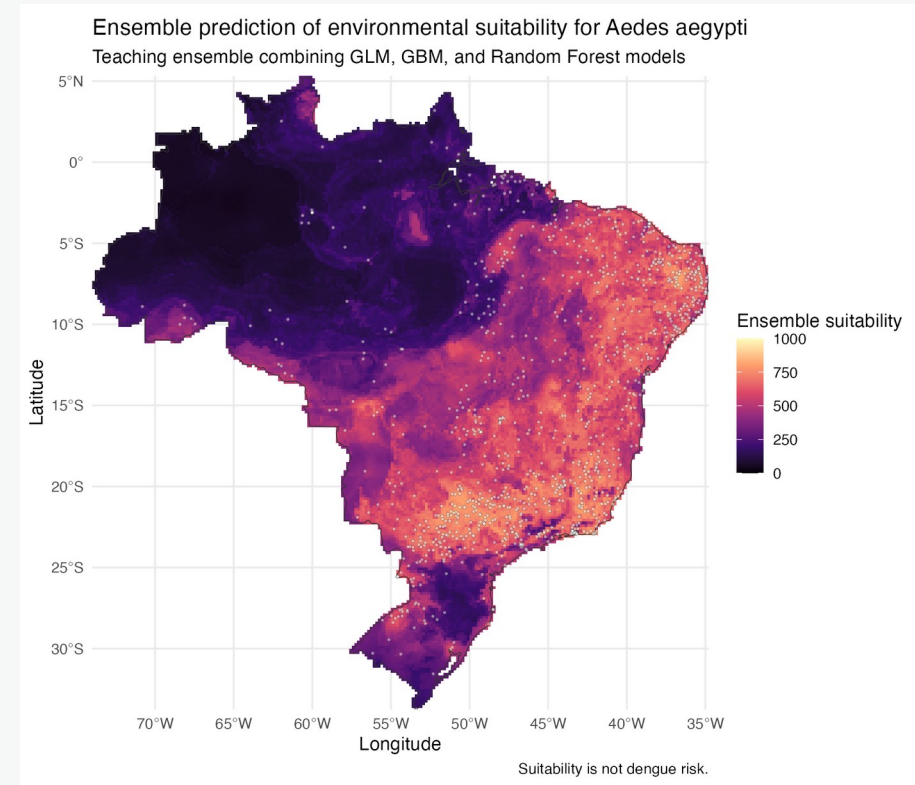


Species distribution models for dengue vector mapping

A beginner-friendly R workshop using *Aedes aegypti*, GBIF, Brazil climate data, and biomod2

90 min guided modelling tutorial



The public health question is simple; the model question is narrower

Where could environmental conditions support the vector?

- Dengue transmission needs more than a mosquito.
- A vector SDM estimates environmental suitability for occurrence.
- The map is a hypothesis to inspect, not a final risk product.

Vector presence

Can the vector live here?



Dengue risk

Also needs people, virus, immunity, surveillance, and interventions.

A species distribution is shaped by more than climate

For *Aedes aegypti*, we usually observe a pattern and then ask which processes might explain it.

Abiotic

Temperature
Rainfall
Seasonality
Urban
microclimate

Biotic

Humans as hosts
Larval habitats
Competitors
Predators

Movement

Dispersal
Trade routes
Urban connectivity
Introductions

Observation

GBIF effort
Misidentification
Duplicate records
Coordinate errors

SDMs are useful because they make one part of this story explicit: the association between known occurrences and environmental predictors.

Today we turn observations into a map through one workflow



A good SDM workflow is less about one magic algorithm and more about explicit choices at every step.

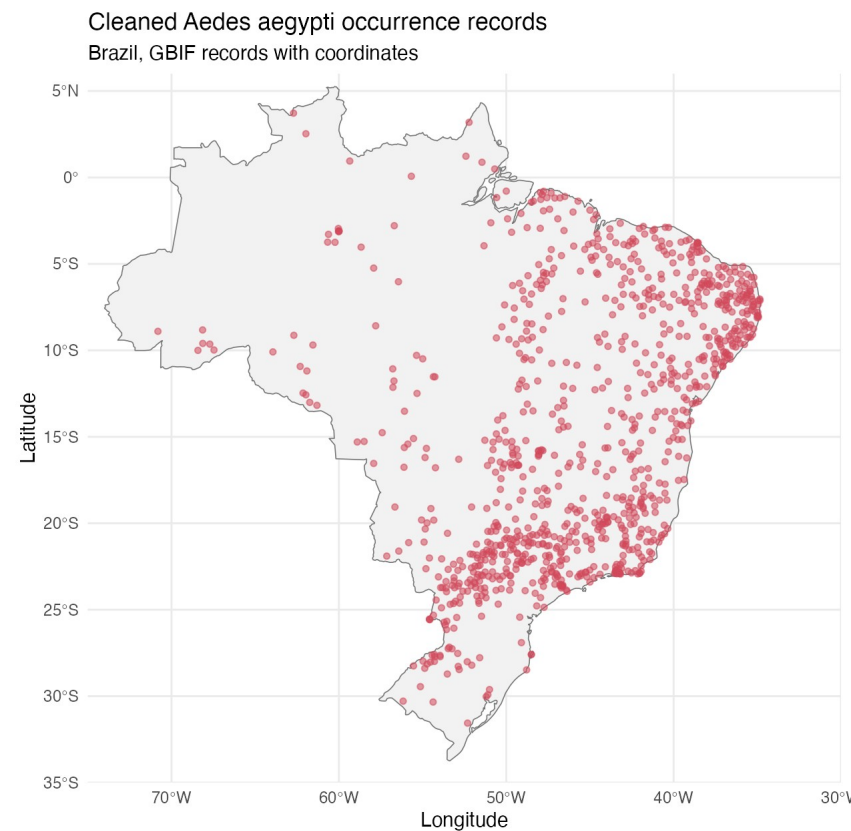
GBIF is a starting point, not a clean truth table

Useful

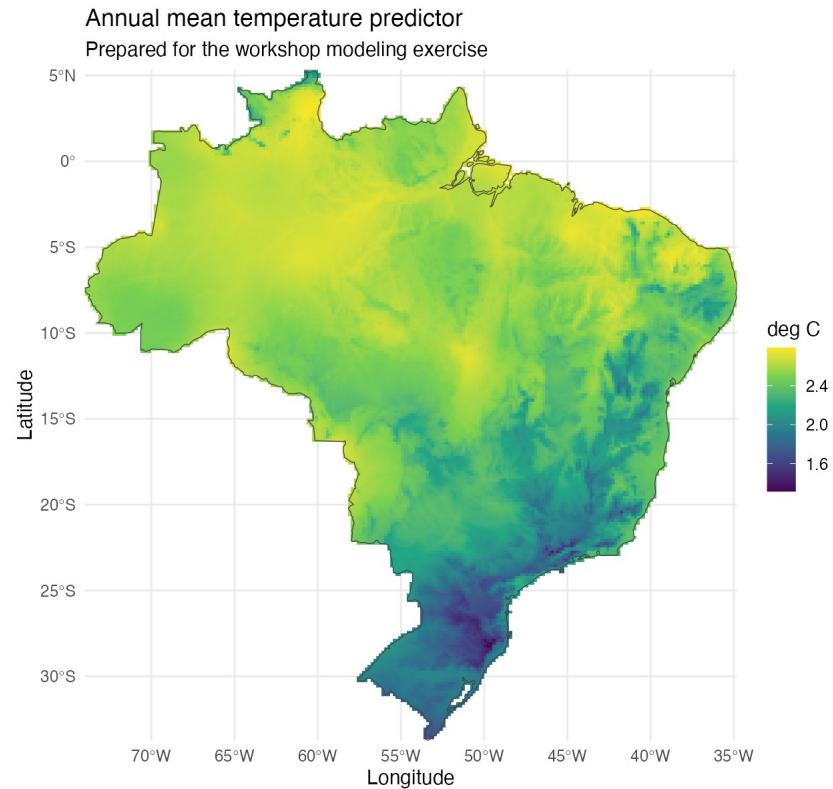
- Open biodiversity records
- Reusable for future species
- Transparent data source

Needs cleaning

- Missing coordinates
- Duplicate points
- Points outside the study area
- Sampling bias



Predictors describe the environmental stage



In the workshop we keep four climate predictors so the model runs quickly and the interpretation stays readable.

Annual mean temperature

Temperature seasonality

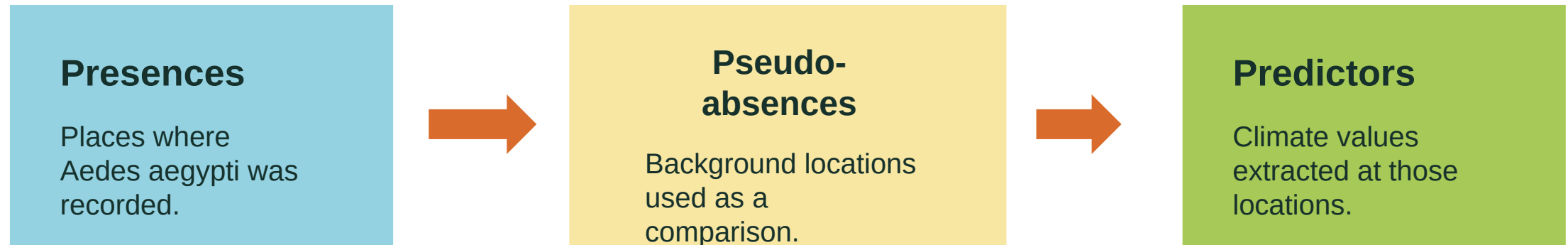
Annual precipitation

Precipitation seasonality

Question for the room: which of these might matter most for a container-breeding mosquito?

Presence-only data need a comparison set

BIOMOD_FormatingData() gathers the ingredients into one modelling object.



Pseudo-absences are modelling choices, not confirmed places where the mosquito is absent.

Different algorithms ask different shapes of question

GLM

Fits a simple statistical relationship;
transparent and interpretable.

smooth trend

GBM

Builds many small trees in sequence;
flexible, often predictive.

boosted trees

RF

Builds many trees on random subsets;
robust, less transparent.

many trees vote

This is why we run more than one model: disagreement is information.

Evaluation asks whether predictions generalise



TSS

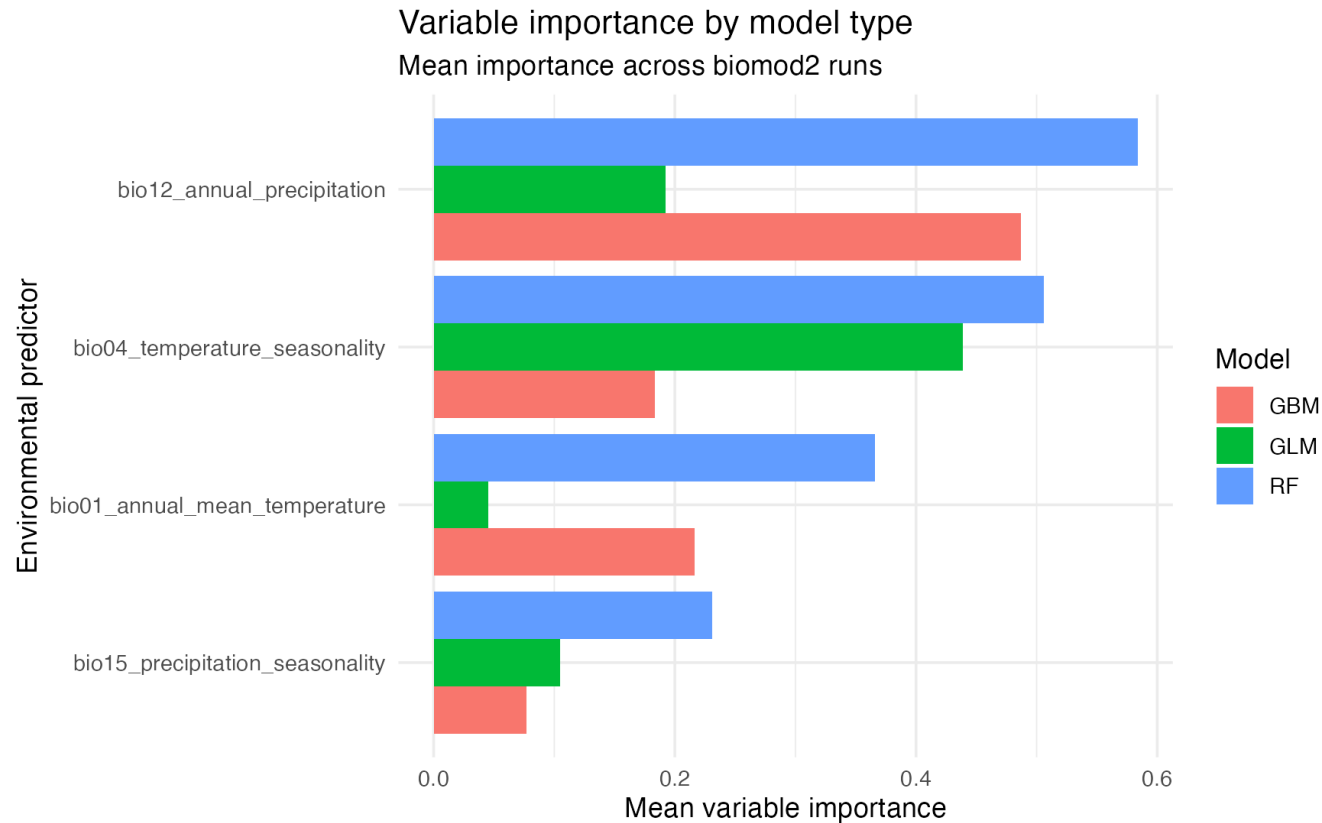
Balances sensitivity and specificity after choosing a threshold.

AUCroc

Measures ranking ability across possible thresholds.

A high score does not remove the need to inspect the map.

Variable importance helps explain what the models used

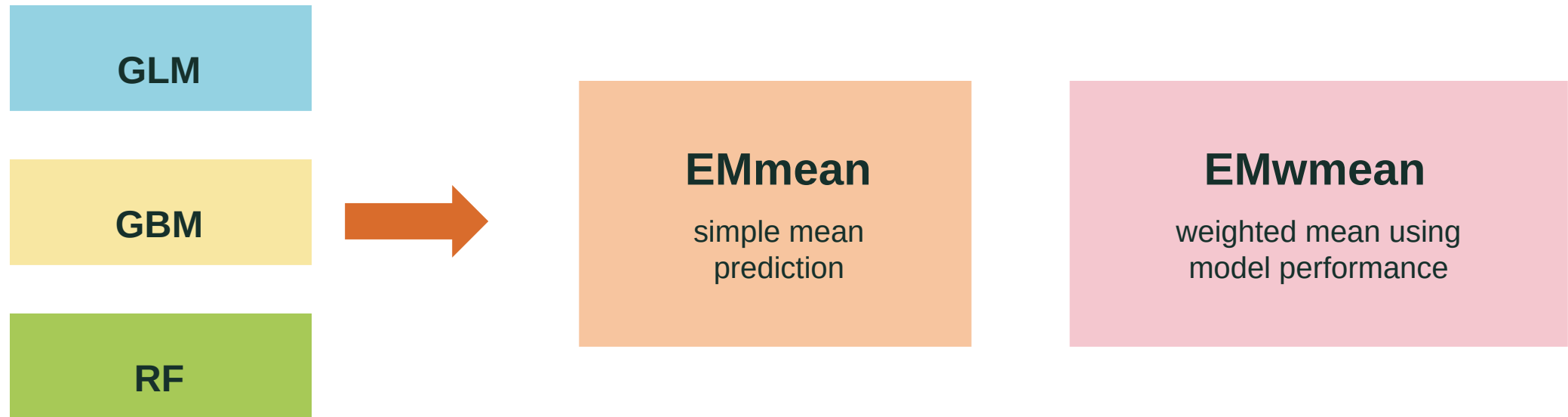


Interpret with care

- Importance is model sensitivity, not biological proof.
- Different algorithms may rank predictors differently.
- Use it to start questions about ecology and sampling.

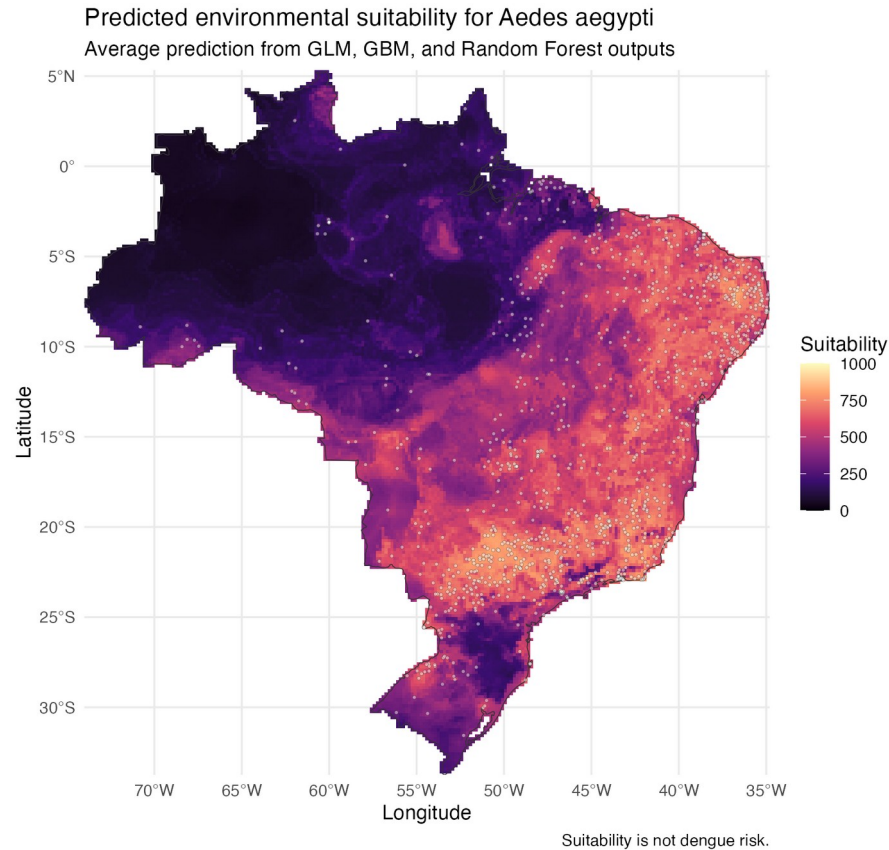
Ensemble modelling keeps the model disagreement visible

We do not need to choose only one model.

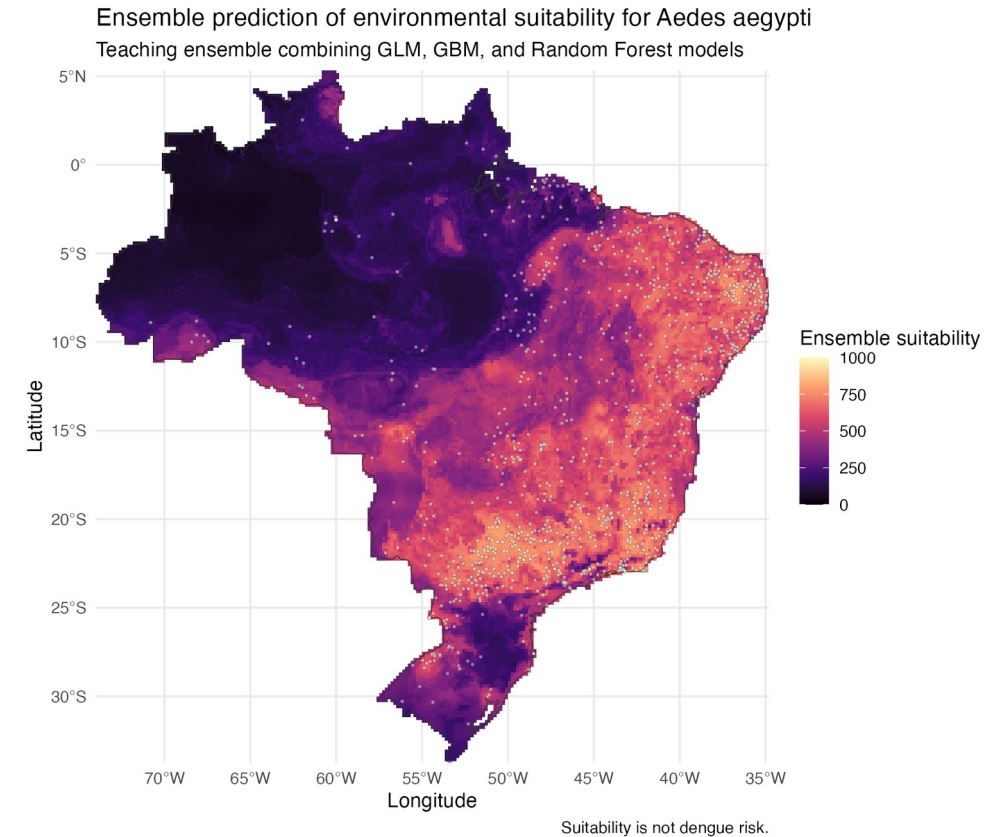


The ensemble is a summary of model behaviour, not a guarantee of truth.

Projection turns fitted models into Brazil maps

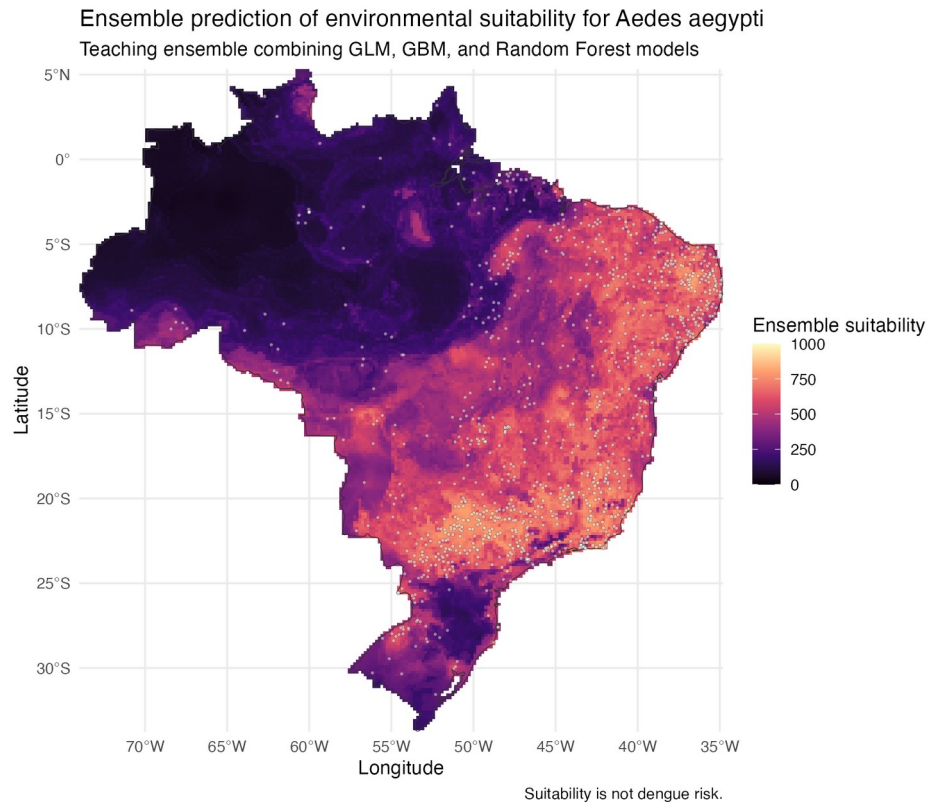


Individual models averaged



Ensemble projection

Read the map as a hypothesis to interrogate



Ask before acting

- Where is suitability high?
- Where are occurrence records sparse?
- What could be climate signal?
- What could be sampling bias?
- What dengue data would change the interpretation?

Your job today is not to become an SDM expert

It is to leave with a working R script, a reusable modelling workflow,
and sharper questions for infectious disease applications.

Run the code

Inspect choices

Question maps

Now let us build the model line by line.